

MUHAMMAD FAISAL

Electrical Engineer | BMS & Automation | Embedded Systems | Solar & Power Systems

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PROFESSIONAL SUMMARY

Results-driven Electrical Engineer with 6+ years of progressive experience in Building Management Systems (BMS), industrial automation, and embedded systems across Saudi Arabia's premier infrastructure projects including NEOM, KAFD, and Riyadh Metro. Professional with proven expertise in Siemens Desigo CC, SCADA integration, Modbus/TCP/IP, and IoT-enabled monitoring solutions. Successfully Developed & delivered Pakistan's first EV charging infrastructure and solar-BMS integration for NEOM's renewable energy initiative. Fluent in English with working knowledge of Arabic. Committed to delivering Saudi Vision 2030-aligned smart city and sustainability solutions.

Core Competencies:

- BMS Integration & Commissioning (Siemens Desigo CC, DXR/PXC, TXM)
- Industrial Protocols (Modbus RTU/TCP, RS485, CAN Bus, BACnet, Lon, KNX)
- Embedded Systems (Atmel, STM32, ESP32, C/C++, Python, Linux)
- PCB Design (Altium, Eagle - up to 6-layer)
- HMI/GUI Development (Nextion, TJC)
- SCADA Integration (Allen Bradley, Siemens)
- Solar PV Systems & Battery Management Systems

PROFESSIONAL EXPERIENCE

SIEMENS | Riyadh, Saudi Arabia

Electrical Engineer - BMS & Automation | Nov 2023 – May 2026

Granada Business Park

- Orchestrated BMS upgrade initiative affecting 11 commercial High rise & Low rise buildings, replacing 105+ outdated controllers.
- Engineered integration strategies for Siemens Desigo CC monitoring platform, improving fault detection by 40%.
- Conducted on-site commissioning of DXRs, PXC's, and field devices across multiple buildings.

- Reduced maintenance response time by 50% through enhanced system diagnostics and reporting.
- Collaborated with 3 subcontractor teams to ensure seamless automation system integration.

Riyadh Metro Line 3 - Power Shelters

- Integrated Allen Bradley SCADA with Siemens BMS across 5+ power shelters, enabling unified monitoring and control of critical infrastructure.
- Engineered automated HVAC control logic reducing manual intervention by 65% and improving system response time to <2 seconds.
- Implemented Modbus RTU/TCP integration for field sensors, achieving 99.5% data acquisition accuracy.
- Developed auto/manual control selection interfaces from SCADA to BMS, enhancing operator flexibility.
- Reduced energy consumption by 18% through optimized HVAC scheduling algorithms.

Saudi National Bank (SNB) Headquarters - KAFD

- Led BMS site operations and system upgrades for 41-story high-rise facility in Riyadh's financial district.
- Programmed and commissioned Siemens DXR & PXC controllers for HVAC and lighting control.
- Upgraded legacy controllers and thermostats using Siemens QMX3-P34 and P44, TXM modules and Desigo CC, increasing system reliability by 35%.
- Managed 13-person team including subcontractors, ensuring 100% on-time delivery of automation milestones.
- Provided 24/7 technical support for electrical, HVAC, and automation systems, maintaining 99.9% uptime during business hours.
- Coordinated directly with client stakeholders to customize BMS interfaces for facility management needs.

NEOM – Robot & Solar BMS Integration

- Designed and implemented custom integration Electronics circuit between Robot shelter, solar inverter systems and Siemens DDC controllers via Modbus TCP/IP.
- Developed firmware and circuit design enabling real-time solar analytics through BMS dashboards.
- Led remote site support and commissioning across 13 NEOM zones, establishing comprehensive solar monitoring.
- Achieved 100% successful data transmission from 13+ solar sites to centralized BMS platform.

TESLA INDUSTRIES | Islamabad, Pakistan

Electrical Engineer - Embedded Systems & Power Electronics | Jan 2020 - Oct 2023

EVX AC Charger (Pakistan's First EV AC Charger)

- Led end-to-end design (hardware + firmware) for 7kW single-phase and 22kW three-phase EV chargers.
- Integrated smart features including app-based control, dynamic power adjustment (1-7kW / 3.6-22kW), and peak tariff scheduling.
- Achieved 97.5% charging efficiency, exceeding industry benchmarks.
- Engineered App & web-enabled interface for remote monitoring and control.

EZ-Swap EV Charging Stations

- Designed modular charging docks for Pakistan's first E-Bike sharing system.
- Developed Hardware & firmware for battery identification and authentication protocols.
- Implemented MOD Bus RTU communication for battery management system integration.

30kW DC Power Supply

- Built high-voltage DC supply system with configurable output for industrial HVDC laboratories.
- Developed GUI and firmware using Nextion HMI for precise voltage/current control.
- Achieved $\pm 0.5\%$ voltage regulation across full operating range.

Zero Export Manager

- Engineered control devices prevent reverse current flow into grid from on-grid solar inverters.
- Designed 4-layer PCB using Eagle for industrial-grade reliability.
- Reduced utility penalties for clients by eliminating reverse feed-in charges.

Solar Inverter BMS Integration

- Programmed lithium battery BMS data integration via RS485 into solar inverter UI.
- Enabled seamless monitoring of battery SOC, temperature, and cell balancing parameters.

EDUCATION

Bachelor of Science in Electrical Engineering & Technology | Sep 2018 - Sep 2022

University of Sargodha, Pakistan

Diploma of Associate Engineer (DAE) in Electronics | Aug 2015 - Aug 2018

HIRA College of Technology, Pakistan